

Menu-Driven Array Operations in C

Counting Elements, Linear Search and Merging Arrays

Aim

To write a menu-driven C program using switch-case and arrays to count even, odd, positive, negative and zero elements, search for an element using linear search, merge two arrays, and exit the program.

Theory / How the Program Works

The program repeatedly displays a menu using a **do-while loop**. A **switch statement** calls a separate void function for each array operation. In the counting operation, every element is tested for parity and sign. Linear search checks the array from the first element to the last until the required value is found. Merging copies all elements of the first array followed by all elements of the second array into a third array. Array sizes are validated before use.

Menu

Choice	Operation
1	Count even, odd, positive, negative and zero elements
2	Search for an element using linear search
3	Merge two arrays
4	Exit

Algorithm

1. Start and declare the menu choice.
2. Display the menu and read the user's choice.
3. Use switch-case to perform the selected operation.
4. For choice 1, read and validate the array size, input the elements, and initialize five counters to zero.
5. Traverse the array. Use element % 2 to count even and odd values. Use comparisons with zero to count positive, negative and zero values. Display all five counts.
6. For choice 2, read the array and the search key. Compare the key with each element from left to right. If a match is found, display its position; otherwise display 'not found'.
7. For choice 3, read and validate the sizes of two arrays and input their elements. Copy the first array into a third array, then append the second array. Display the merged array.
8. For choice 4, display an exit message.
9. For any other menu choice, display an appropriate error message.
10. Repeat the menu until choice 4 is selected.
11. Stop.

Test-Case Table

Test	Operation / Input	Expected result
1	Count: [-4, 0, 3, 8, -1, 2]	Even=4, Odd=2, Positive=3, Negative=2, Zero=1
2	Count: [0, 0, 0]	Even=3, Odd=0, Positive=0, Negative=0, Zero=3
3	Search: [10,20,30,40,50], key=30	30 found at position 3
4	Search: [2,4,6,8], key=5	5 not found in the array

Test	Operation / Input	Expected result
5	Merge: [1,2,3] and [4,5]	Merged array: 1 2 3 4 5
6	Array size = 0, then valid size	Invalid size is rejected and input is requested again
7	Menu choice = 9	Invalid-choice message; menu is displayed again
8	Menu choice = 4	Program exits normally

C Program

```
#include <stdio.h>

#define MAX 100

void countElements(void);
void linearSearch(void);
void mergeArrays(void);

int main(void)
{
    int choice;
    do {
        printf("\n===== ARRAY OPERATIONS MENU =====\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: countElements(); break;
            case 2: linearSearch(); break;
            case 3: mergeArrays(); break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid choice! Enter 1 to 4.\n");
        }
    } while (choice != 4);
    return 0;
}

void countElements(void)
{
    int a[MAX], n, i;
    int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;
    do {
        printf("Enter array size (1-%d): ", MAX);
        scanf("%d", &n);
        if (n <= 0 || n > MAX) printf("Invalid size!\n");
    } while (n <= 0 || n > MAX);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) scanf("%d", &a[i]);

    for (i = 0; i < n; i++) {
        if (a[i] % 2 == 0) even++; else odd++;
        if (a[i] > 0) positive++;
        else if (a[i] < 0) negative++;
        else zero++;
    }
    printf("Even = %d\nOdd = %d\nPositive = %d\nNegative = %d\nZero = %d\n",
        even, odd, positive, negative, zero);
}

void linearSearch(void)
{
    int a[MAX], n, i, key, found = 0;
    do {
        printf("Enter array size (1-%d): ", MAX);
        scanf("%d", &n);
        if (n <= 0 || n > MAX) printf("Invalid size!\n");
    } while (n <= 0 || n > MAX);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) scanf("%d", &a[i]);
    printf("Enter element to search: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++) {
        if (a[i] == key) {
            printf("%d found at position %d.\n", key, i + 1);
            found = 1;
            break;
        }
    }
    if (!found) printf("%d not found in the array.\n", key);
}

void mergeArrays(void)
{
    int a[MAX], b[MAX], c[2 * MAX];
    int n1, n2, i, k = 0;
    do {
        printf("Enter size of first array (1-%d): ", MAX);
```

```

        scanf("%d", &n1);
        if (n1 <= 0 || n1 > MAX) printf("Invalid size!\n");
    } while (n1 <= 0 || n1 > MAX);

    do {
        printf("Enter size of second array (1-%d): ", MAX);
        scanf("%d", &n2);
        if (n2 <= 0 || n2 > MAX) printf("Invalid size!\n");
    } while (n2 <= 0 || n2 > MAX);

    printf("Enter %d elements of first array: ", n1);
    for (i = 0; i < n1; i++) scanf("%d", &a[i]);
    printf("Enter %d elements of second array: ", n2);
    for (i = 0; i < n2; i++) scanf("%d", &b[i]);

    for (i = 0; i < n1; i++) c[k++] = a[i];
    for (i = 0; i < n2; i++) c[k++] = b[i];

    printf("Merged array: ");
    for (i = 0; i < k; i++) printf("%d ", c[i]);
    printf("\n");
}

```

Compilation and Execution

The program was compiled using **gcc -Wall -Wextra array_menu.c -o array_menu**. Compilation completed successfully. It was executed to test counting, successful linear search, merging, an invalid menu choice, and Exit.

Sample Output

```
===== ARRAY OPERATIONS MENU =====
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: Enter array size (1-100): Enter 6 elements: Even = 4
Odd = 2
Positive = 3
Negative = 2
Zero = 1

===== ARRAY OPERATIONS MENU =====
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: Enter array size (1-100): Enter 5 elements: Enter element to search: 30 found at position 3.

===== ARRAY OPERATIONS MENU =====
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: Enter size of first array (1-100): Enter size of second array (1-100): Enter 3 elements of first array: Enter 2 elements of second array:

===== ARRAY OPERATIONS MENU =====
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: Invalid choice! Enter 1 to 4.

===== ARRAY OPERATIONS MENU =====
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: Exiting program...
```

Result

The menu-driven C program was written, compiled and executed successfully. It correctly counts array elements by parity and sign, performs linear search, merges two arrays, validates array sizes, handles invalid menu choices, and repeats until Exit is selected.